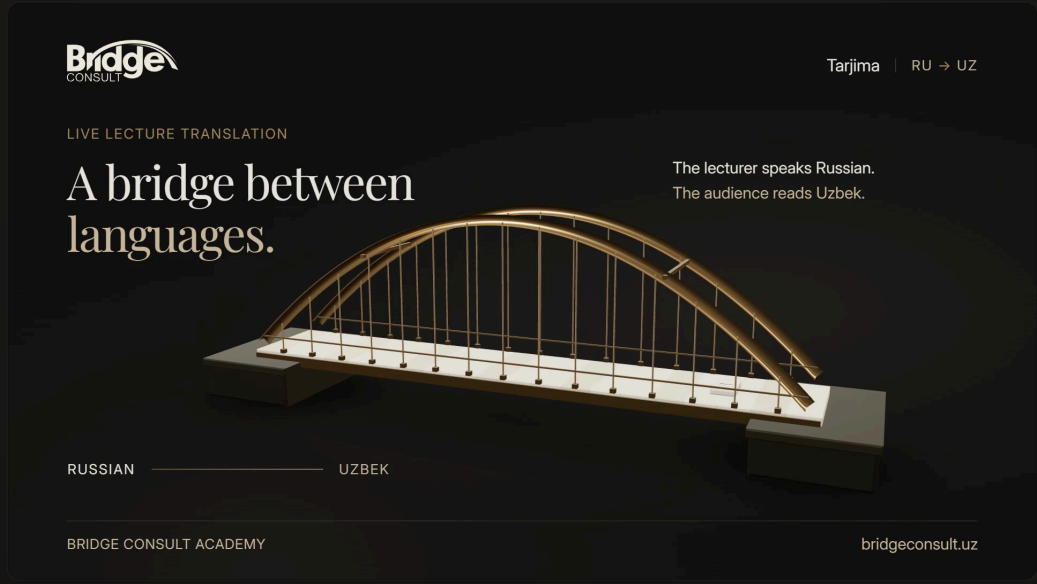
LIVE LECTURE TRANSLATION · RU → UZ

Tarjima. A bridge between languages.

A live translation system for BRIDGE Consult Academy's online courses. The lecturer speaks Russian; attendees read Uzbek subtitles within seconds — in their video call or on their phone, with nothing to install.

CLIENT	BRIDGE Consult Academy, Tashkent
SCOPE	Product design, engineering, terminology, launch
BUILT	12–23 September 2026 · in production
LIVE AT	bridgeconsult.uz/tarjima



1.9 s	$0.0023	92	0	1×	67
median model time per fragment (measured, Sonnet 5)	cost per translated fragment (measured)	curated contract terms in the glossary	installs or sign-ups for attendees	translation per phrase, whatever the audience size	automated checks, run without network or API spend

CONTEXT

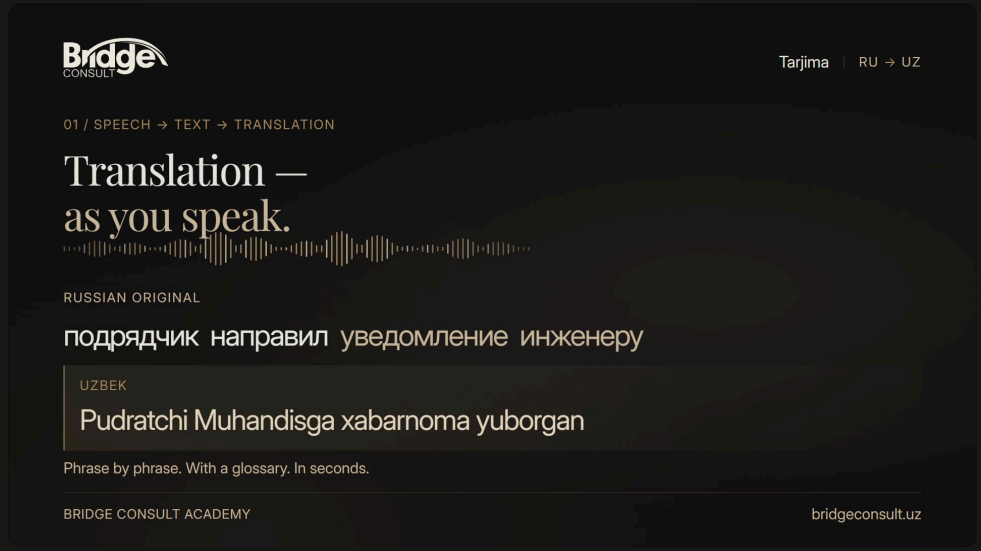
Lectures in Russian. An audience that reads Uzbek.

BRIDGE Consult trains engineers, contract managers and lawyers who work on infrastructure projects in Uzbekistan. The courses are taught in Russian; the programme promises delivery **with translation into Uzbek**.

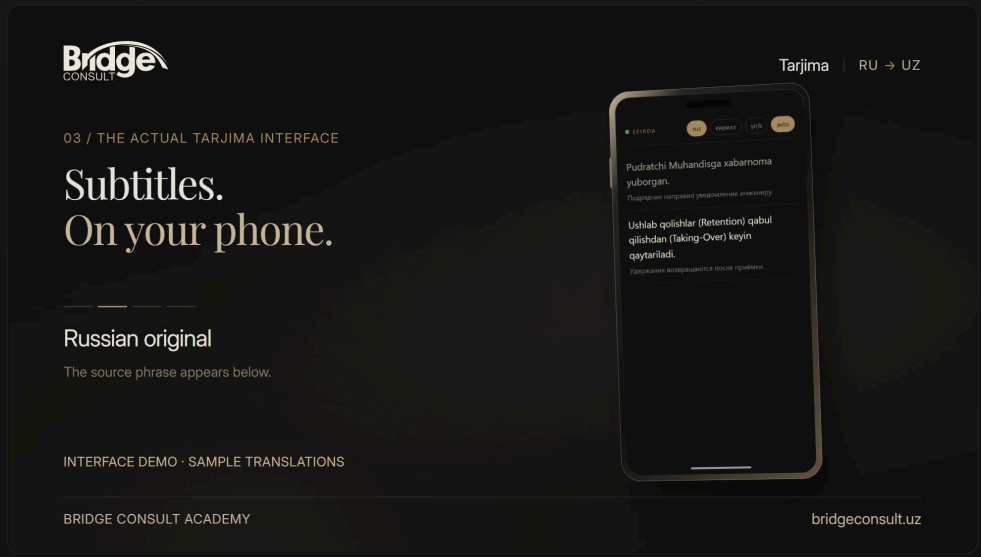
Hiring simultaneous interpreters for every session is expensive, and generic machine translation stumbles exactly where it matters: contract terminology. For much of this vocabulary there is no settled Uzbek legal term, so a generic tool invents a new one each time.

WHAT WE BUILT

- **Lecturer page** — press Start in Chrome and teach as usual. Speech is recognised, cut into short fragments and translated as the lecturer speaks.
- **Subtitles inside the video call** — lines are pushed to Zoom's closed-caption channel; attendees just tap CC.
- **Attendee page** — a link or QR code opens live subtitles on any phone: Latin or Cyrillic Uzbek, large type, Russian original on request.
- **The company's own glossary** — every contract term is rendered the way BRIDGE Consult writes it.



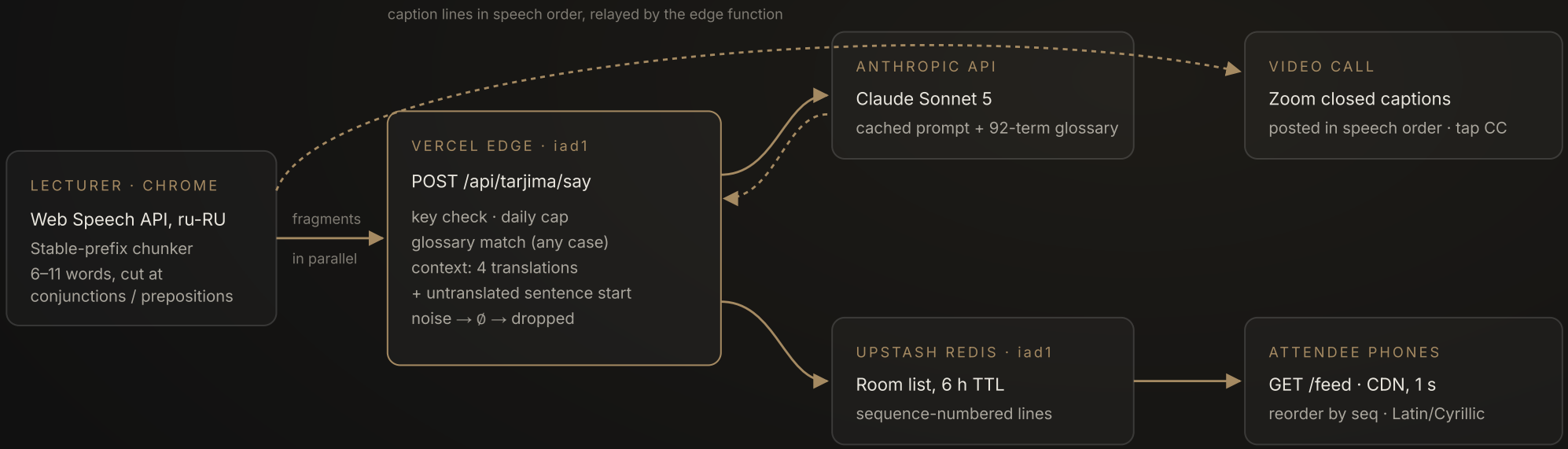
SPEECH → TEXT → TRANSLATION, FRAGMENT BY FRAGMENT



THE ACTUAL ATTENDEE INTERFACE

HOW A SENTENCE TRAVELS

From voice to subtitle in one short hop.



No audio reaches our server

Recognition runs in the lecturer's Chrome; the backend only ever receives text. Transcripts expire from storage after six hours.

One translation, any audience

The feed returns the same response to every attendee, so the CDN serves it from one cache key: storage sees about one read per second whether 10 or 300 people are watching.

Functions pinned next to data

Edge functions run in Washington (iad1), beside the Redis store and the model API, so every internal hop is local. The only long hop is browser → function.

WHAT MAKES IT FAST, ACCURATE AND HONEST

Twelve decisions that matter.

01 · LATENCY

Translate while the lecturer is still talking

Chrome only closes a phrase on a pause. A stable-prefix chunker ships 6–11 words the recogniser has stopped revising, preferring cuts before conjunctions or prepositions, and holds back the last two words.

02 · LATENCY

Parallel, but never out of order

Fragments are translated concurrently. Every line carries a session id and sequence number; attendee pages re-sort, and captions go to Zoom strictly in speech order — a fragment stuck for 5 s never blocks the rest.

03 · ACCURACY

Context across fragments

Each request carries the last four translations as dialogue, plus the not-yet-translated start of the sentence, so a half-sentence is rendered as a continuation rather than a fresh start.

04 · COST · SPEED

Prompt caching by design

Instructions and the full glossary form one invariant ~1,800-token block, cached across the lecture. Anything that changes per fragment lives in the messages; an automated test fails if the cached block ever varies.

05 · ACCURACY

A terminology engine for Russian morphology

Terms are matched in any grammatical case with bounded suffixes («уведомлении», «Инженерами»), longest match first, and without false hits («спор» ≠ «спортивный»).

06 · ACCURACY

Recovering English terms from speech

Recognition writes spoken English terms in Cyrillic — «клейм», «дааб», «иписи». These are detected and restored as Claim, DAAB, IPC; Russian terms stay in the company's Uzbek.

07 · TRUST

Never pass Russian off as Uzbek

If translation fails, attendees see the Russian original, marked in red: “Tarjima ishlamadi”. Noise and filler words are flagged by the model as ∅ and dropped instead of turning into apologies on screen.

08 · INTEGRATION

Zoom captions, validated server-side

The host pastes Zoom's caption API token; only https://*.zoom.us/closedcaption is accepted. Sequence counters are kept per breakout room, with a fallback if the language code is rejected.

09 · SAFETY

Cost and abuse controls

A lecturer key, a daily fragment cap checked before any model call, a whitelist for model overrides, and no secrets in client code (audited).

10 · RESILIENCE

Built for a two-hour live session

Recognition restarts itself and flushes unfinished words; the screen is kept awake; 12 s server / 20 s client timeouts; attendee pages ride out flaky mobile networks and pause polling in background tabs.

11 · ACCESS

Latin or Cyrillic Uzbek

A rule-based transliterator (oʻ→ў, gʻ→ғ, sh→ш, word-initial e→э) lets older engineers read in the script they prefer, while English contract terms stay in Latin.

12 · EVIDENCE

Decisions made by measurement

A benchmark harness replays realistic recogniser output against production, compares models side by side, and writes a blind table for a native-speaker reviewer.

MODEL BENCHMARK · PRODUCTION · 22 SEP 2026

We picked the model with data.

Ten identical lecture fragments, written the way speech recognition returns them, sent through the live system with four model configurations.

MODEL	MODEL TIME, MEDIAN	95TH PCT	PER FRAGMENT	VERDICT
Claude Sonnet 5 DEFAULT	1.9 s	2.3 s	$0.0023	Fluent, uses the glossary
Claude Opus 5	3.1 s	4.6 s	$0.0059	Comparable quality, slower
Opus 5 · fast mode	3.1 s	4.1 s	premium	No real gain here
Claude Haiku 4.5	1.0 s	2.0 s	$0.0026	Grammar errors

Model time measured server-side. Quality judged on this sample by a non-native reviewer; the blind comparison table is prepared for a native legal translator.

A BUG THE BENCHMARK CAUGHT

The first run showed the larger model replacing Uzbek terms with English ones even when the lecturer spoke Russian — «уведомление о претензии» became `Notice of Claim`. The instruction was rewritten, and the re-run confirmed the fix.

REAL OUTPUT, FROM PRODUCTION

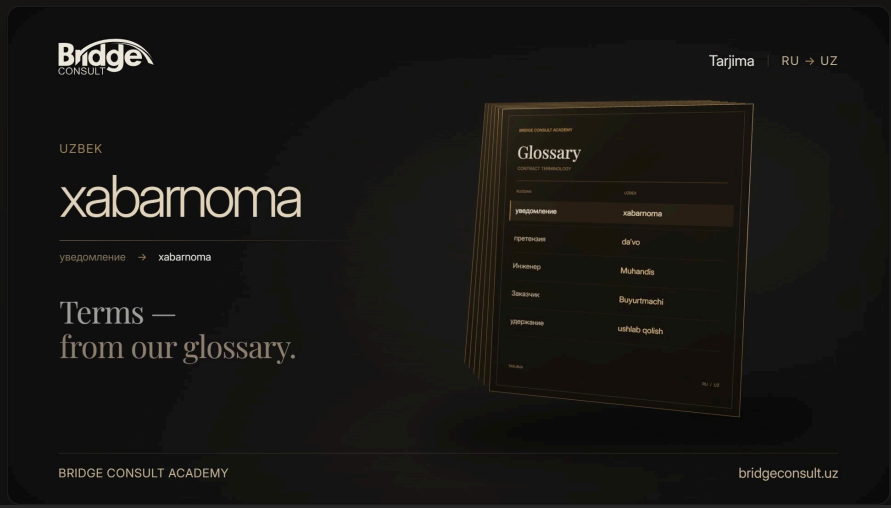
> подрядчик направил уведомление инженеру
> Pudratchi Muhandisga xabarnoma yuborgan

> удержания возвращаются после приёмки
> Ushlab qolishlar (Retention) qabul qilishdan (Taking-Over) keyin qaytariladi

GLOSSARY IN ACTION

уведомление	→	xabarnoma
претензия	→	da’vo
Инженер	→	Muhandis
продление срока	→	muddatni uzaytirish (EOT)
удержание	→	ushlab qolish (Retention)

Every Uzbek term is taken from the company's own published bilingual materials — none invented. Conflicting variants across sources are listed for sign-off by the company's lead expert.



STACK

SPEECH RECOGNITION	Web Speech API in Google Chrome, ru-RU, continuous with interim results
TRANSLATION	Anthropic Claude via the official TypeScript SDK; Sonnet 5 by default, Opus 5 / Haiku 4.5 switchable by one variable; low effort; prompt caching
BACKEND	Vercel Edge Functions pinned to iad1 · `POST /api/tarjima/say` (translate · zoom · reset) · `GET /api/tarjima/feed`
STORAGE	Upstash Redis over REST, pipelined commands, no client library
DELIVERY	Attendee polling every 1 s behind a 1 s shared CDN cache; Zoom closed-caption API
FRONT END	Framework-free HTML, CSS and JavaScript; no build step, no dependencies in the browser
HOST SITE	Astro 6 marketing site, bridgeconsult.uz
CODEBASE	~2,200 lines · 13 commits over 12 days

QUALITY

FUNCTION TESTS	38 checks with fake model, storage and Zoom servers — every branch, no network, no spend (`test.mjs`)
GLOSSARY TESTS	29 cases: inflection, transliterated English, false-positive traps, duplicate detection
BENCHMARK	`eval.mjs` — 30 recogniser-style phrases, multi-model, blind review table

OPERATING PARAMETERS

FRAGMENT SIZE	6–11 words; 2 trailing words held while the recogniser revises
CONTEXT PER REQUEST	4 previous translations + up to 2 pending fragments
GLOSSARY	92 terms, ~1,800-token cached prompt block
TIMEOUTS	12 s server with one retry · 20 s client abort
RETENTION	Transcript lines expire after 6 hours
SAFETY LIMITS	Lecturer key · 5,000 fragments / day · model whitelist
ROOMS	Parallel classes via `?room=` ; one "New session" reset per room
RUNNING COST	≈ $2 per lecture hour for translation (measured per fragment, estimated fragments per hour); storage within free tier

KNOWN LIMITS

- Lecturer must use Chrome — Brave disables speech recognition.
- Roughly 3–4 s from words to subtitle. Uzbek puts the verb last, so a phrase cannot be translated before it is said.
- Glossary pending sign-off by a native-speaking legal expert.